

Clinical Risk Assessment Report: Patient 43

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.2% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.51x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.2%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.32 [Bottom 19%] (+2.5%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.05 [Avg (46th %ile)] (-2.2%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Patient Age (Age) **Value: 67.0 [Avg (64th %ile)] (-1.8%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.25 [Avg (48th %ile)] (-1.2%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 3.3 [Top <1%] (-0.9%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.42 [Bottom <1%] (-0.9%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Carcinoembryonic Antigen (CEA) **Value: 1.98 [Avg (26th %ile)] (-0.8%)**
Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.08 [Avg (41th %ile)] (-0.8%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Machine Learning Feature Attribution (SHAP Decision Path)

